EED 101 INTRODUCTION TO ELECTRICAL ENGINEERING

SPRING 2017 **INSTRUCTOR: DR G NAVEEN BABU**

PROBLEM SET 6

REFERENCE TEXT BOOK: INTRODUCTORY CIRCUIT ANALYSIS, ROBERT BOYLESTAD, ***12TH EDITION***

Chapter 13: 19,23,33,35,40,41,50,53

Chapter 14: 20,31,35

23. $6 \times 10^{-3} = 30 \times 10^{-3} \sin \alpha$

$0.2 = \sin \alpha$

$\alpha = \sin^{-1} 0.2 = \mathbf{11.54°}$ and $180° - 11.54° = \mathbf{168.46°}$

33. $$\left.\begin{aligned} v &= 2 \sin(\omega t - \underbrace{30° + 90°}_{+60°}) \\ i &= 5 \sin(\omega t + 60°) \end{aligned}\right\} \textbf{ in phase}$$

35. $$T = \frac{1}{f} = \frac{1}{1000\text{ Hz}} = 1\text{ ms}$$

$$t_1 = \frac{120°}{180°}\left(\frac{T}{2}\right) = \frac{2}{3}\left(\frac{1\text{ ms}}{2}\right) = \mathbf{\frac{1}{3}\ ms}$$

40. $$G = \frac{\left[\frac{1}{2}(4\text{ ms})(20\text{ mA})\right] - (2\text{ ms})(8\text{ mA}) - \frac{1}{2}(2\text{ ms})(8\text{ mA})}{8\text{ ms}}$$

$$= \frac{40\text{ mA} - 16\text{ mA} - 8\text{ mA}}{8} = \frac{16\text{ mA}}{8}$$

$$= \mathbf{2\ mA}$$

41. $$G = \frac{(35\text{ V})(5\text{ ms}) - \frac{1}{2}(20\text{ V})(20\text{ ms}) - (20\text{ V})(15\text{ ms}) - \frac{1}{2}(20\text{ V})(7.5\text{ ms}) + \frac{1}{2}(20\text{ V})(7.5\text{ ms}) + \frac{1}{2}(20\text{ V})(15\text{ ms}) + 0}{75\text{ ms}}$$

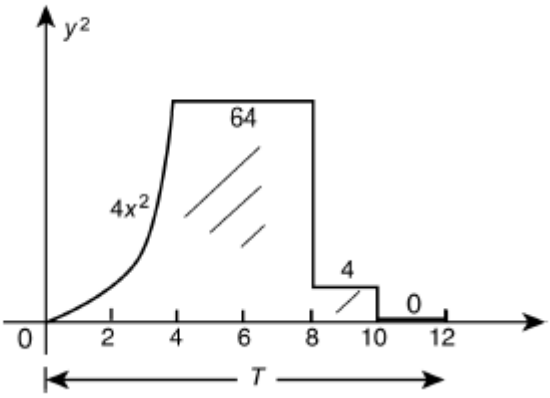
$$= \frac{175\text{ V} - 200\text{ V} - 300\text{ V} - 75\text{ V} + 75\text{ V} + 150\text{ V}}{75}$$

$$= \frac{400\text{ V} - 575\text{ V}}{75} = \mathbf{2.33\text{ V}}$$

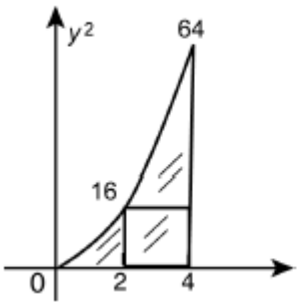
50. $$V_{rms} = \sqrt{\frac{(3\text{ V})^2(2\text{ s}) + (2\text{ V})^2(2\text{ s}) + 0 + (-1\text{ V})^2(2\text{ s}) + (-3\text{ V})^2(2\text{ s}) + (-2\text{ V})^2(2\text{ s})}{12\text{ s}}}$$

$$= \sqrt{\frac{54}{12}\text{ V}^2} = \sqrt{4.5\text{ V}^2} = \mathbf{2.12\ V}$$

53. a.



b.



$$A_1 = \frac{1}{2}(2)(16) + (2)(16) + \frac{1}{2}(2)(48) = 96$$

$$\text{Area} = 96 + (4)(64) + (2)(4) = 96 + 256 + 8 = \mathbf{360}$$

c. $$\text{rms} = \sqrt{\frac{360}{12}} = \sqrt{30} = \mathbf{5.48}$$

d. $$G = \frac{\frac{1}{2}(4)(8) + 4(8) - 2(2)}{12} = \frac{16 + 32 - 4}{12} = \mathbf{3.67}$$

e. rms ≅ **1.5 (average value)**

Chapter 14:

20. a. υ leads i by 90° $\Rightarrow L$, $X_L = V_m/I_m = 550\text{ V}/11\text{ A} = 50\ \Omega$

$$L = \frac{X_L}{\omega} = \frac{50\ \Omega}{377\text{ rad/s}} = \mathbf{132.63\ mH}$$

b. υ leads i by 90° $\Rightarrow L$, $X_L = V_m/I_m = 36\text{ V}/4\text{ A} = 9\ \Omega$

$$L = \frac{1}{\omega X_L} = \frac{1}{(754\text{ rad/s})(9\ \Omega)} = \mathbf{147.36\ \mu H}$$

c. υ and i are in phase $\Rightarrow R$

$$R = \frac{V_m}{I_m} = \frac{10.5\text{ V}}{1.5\text{ A}} = \mathbf{7\ \Omega}$$

31. $$P = \frac{V_m I_m}{2}\cos\theta$$

$$500\text{ W} = \frac{(50\text{ V})I_m}{2}\ (0.5) \Rightarrow I_m = 40\text{ A}$$

$$i = \mathbf{40\ \sin(\omega t - 50°)}$$

35. a. $$X_{C_1} = \frac{1}{2\pi f C_1} = \frac{1}{\omega C_1} = \frac{1}{(10^4\text{ rad/s})(2\ \mu\text{F})} = 50\ \Omega$$

$$X_{C_2} = \frac{1}{\omega C_2} = \frac{1}{(10^4)(10\ \mu F)} = 10\ \Omega$$

$$\mathbf{E} = 84.85\text{ V}\angle 60° \qquad \mathbf{I}_1 = \frac{\mathbf{E}}{\mathbf{Z}_{C_1}} = \frac{84.85\text{ V}\angle 60°}{50\ \Omega\angle -90°} = 1.697\text{ A}\angle 150°$$

$$\mathbf{I}_2 = \frac{\mathbf{E}}{\mathbf{Z}_{C_2}} = \frac{84.85\text{ V}\angle 60°}{10\ \Omega\angle -90°} = 8.485\text{ A}\angle 150°$$

$$i_1 = \mathbf{2.4\ \sin(10^4 t + 150°)}$$
$$i_2 = \mathbf{12\ \sin(10^4 t + 150°)}$$

b. $$C_T = 2\ \mu\text{F} \,||\, 10\ \mu\text{F} = 12\ \mu\text{F}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{(10^4\text{ rad/s})(12\ \mu\text{F})}$$
$$= 8.33\ \Omega$$

$$\mathbf{I}_s = \frac{\mathbf{E}}{\mathbf{X}_{C_T}} = \frac{84.85\angle 60°}{8.33\ \Omega\angle -90°}$$
$$= 10.19\text{ A}\angle 150°$$

$$i_s = \mathbf{14.4\ \sin\ (10^4 t + 150°)}$$